

Shin Dong-Yeon

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D., Major: Electrical Engineering., Advisor: Tae-Hyun Oh

Mar. 2026 -

Daejeon, South Korea

Pohang University of Science and Technology (POSTECH)

Master of Science., Major: Electrical Engineering., Advisor: Tae-Hyun Oh

Feb. 2024 - Feb. 2026

Pohang, South Korea

Pohang University of Science and Technology (POSTECH)

Bachelor of Science., Major: Electrical Engineering., Advisor: Chulhong Kim

Feb. 2019 - Feb. 2024

Pohang, South Korea

EXPERIENCE

POSTECH, Advanced Machine Intelligence Lab

Undergraduate Research Internship; Advisor: Tae-Hyun Oh

Jan. - Feb., Jul. - Nov. 2023

Pohang, South Korea

- Research on robust optical flow estimation against exposure change.

SAMSUNG Mobile eXperience, Communication Processor Lab

Undergraduate Research Internship; Advisor: Yongsang Cho

Jul. - Aug. 2022

Suwon, South Korea

- Research on smartwatch-based hand gesture recognition.

POSTECH, Wireless Communication Machine Learning Lab

Undergraduate Research Internship; Advisor: Yo-Seb Jeon

Dec. 2021 - Feb. 2022

Pohang, South Korea

- Research on MIMO system.

SK Hynix, Test Prior Technology Team

Undergraduate Internship

Jun. - Aug. 2021

Seongnam, South Korea

- Research on performance and quality assessment methodologies for SSD and HDD.

AWARD AND HONOR

Outstanding Poster Awards, IPIU 2025

"HDR-NSFF: High Dynamic Range Neural Scene Flow Fields".

Jan. 2025

PUBLICATION

Conference

- [C5] Yejin Yeo, Baek Sung-Eun, **Shin Dong-Yeon**, Lee Jung-Mok, Tae-Hyun Oh, "TypoErase: Training-Free Image Level Defense against Typographic Attacks," *under review*.
- [C4] GeonU Kim, **Shin Dong-Yeon**, Tae-Hyun Oh, "Reflection-aware Generative Novel View Synthesis," *under review*.
- [C3] **Shin Dong-Yeon**, Kim Jun-Seong, Kwon Byung-Ki, Tae-Hyun Oh, "HDR-NSFF: High Dynamic Range Neural Scene Flow Fields," International Conference on Learning Representations (ICLR) 2026.
- [C2] **Shin Dong-Yeon**, Kim Jun-Seong, Kwon Byung-Ki, Tae-Hyun Oh, "Dynamic HDR Radiance Fields via Neural Scene Flow," ICCV Workshop on Wild3D: 3D Modeling, Reconstruction, and Generation in the Wild, 2025.
- [C1] **Shin Dong-Yeon**, Kim Jun-Seong, Kwon Byung-Ki, Tae-Hyun Oh, "HDR-NSFF: High Dynamic Range Neural Scene Flow Fields," Workshop on Image Processing and Image Understanding (IPIU) 2025. ([Outstanding Poster Awards.](#))

PROJECTS

Development of spatio-temporal dynamic 3D cognitive map and lightweight localization software for outdoor autonomous driving environments. *Apr. - Nov. 2025*

- Constructing spatio-temporal cognitive maps from time-varying multimodal data and integrating them with Vision-Language Models to generate semantic descriptions.
- Funded by ETRI, Korea.

Reconstructing 3D cognitive map for driving environments using Vision-Language Models. *Apr. - Nov. 2024*

- Designing efficient techniques for building and leveraging multi-modal cognitive maps.
- Funded by ETRI, Korea.

Robust optical flow estimation against exposure change. *Jul. - Nov. 2023*

- Undergraduate Research Program, POSTECH AMILab.

Smart Watch-Based Gesture Recognition Using Time Series Classification. *Jul. - Aug. 2022*

- Research on recognizing and classifying user hand gestures using PPG and IMU signals from smartwatches.
- Undergraduate Internship, SAMSUNG MX business.

AI Algorithm Development for Estimating State of Lithium-Ion Battery. *Aug. 2021 - Jun. 2022*

- Developed a model to estimate SoC and SoH from battery time-series data.
- Undergraduate thesis project.

TEACHING EXPERIENCE

EECE372 Microprocessor Architecture and Applications *2024*
Serve as a teaching assistant. POSTECH

NAVER Boostcamp AI Tech 7th *2024*
Serve as a teaching assistant for Computer Vision Theory course. NAVER & Upstage

Student Mentoring Program (SMP) *2021 - 2022*
Serve as a tutor for University Physics 1, 2. POSTECH